

ME:504

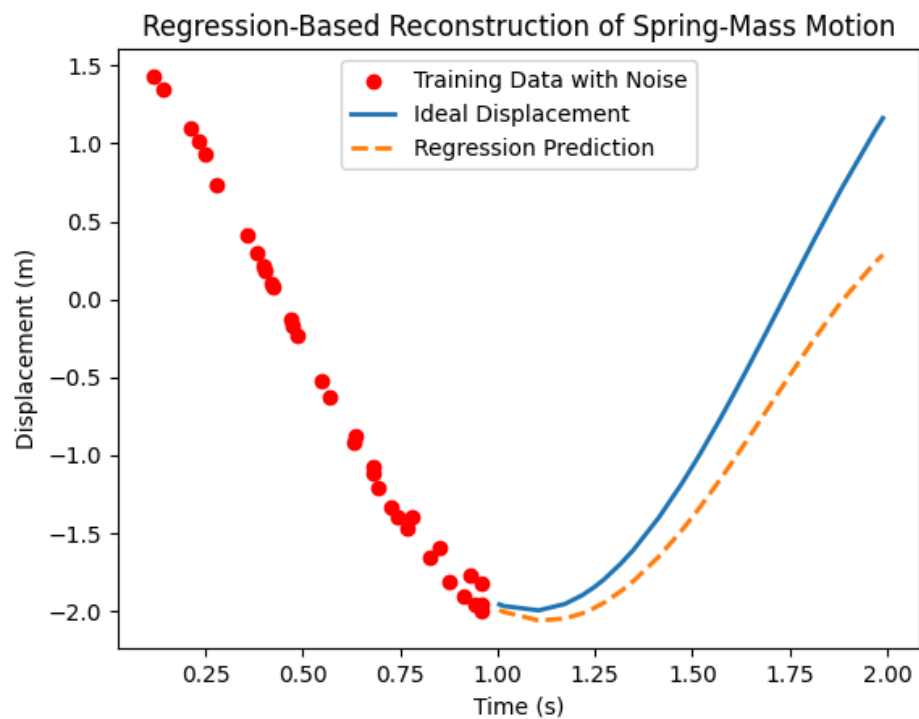
Assignment 2: Result Report

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Predicted Output using the Polynomial Regression Algorithm

Here, degree of polynomial = 4

Noise = $\pm 5\%$



Plot of Training and Testing Error versus degree of polynomial:

Noise = 5% of ideal displacement

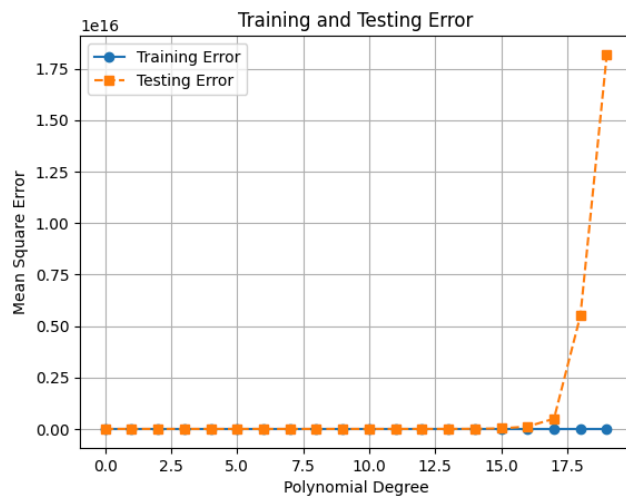


From the above plot,

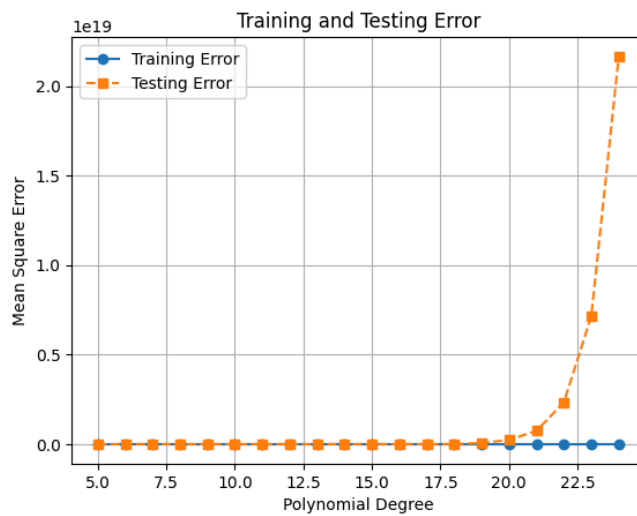
- Capacity of Polynomial = 17.5
- For degree < 17.5, the region is considered as underfitting
- For degree > 17.5, the region is considered as overfitting

To Observe the Effect of Noise on Training and Testing Error

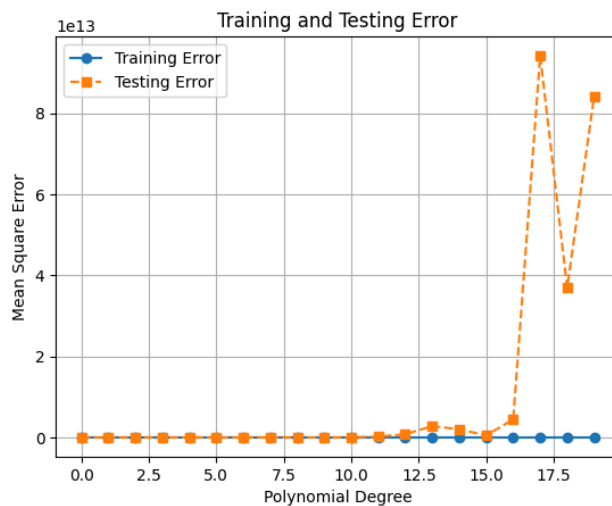
For Noise = 2%



For Noise = 5%



For Noise = 10%



Observations:

- On increasing the noise, the difference between Training and Testing error becomes more and more volatile. We are increasing the randomness in the data.
- Overfitting starts at a lower degree as the value of noise increases.

To Observe the Effect of Data Size on Training and Testing Error

Considering

Noise = 10%

And degree of polynomial = 15 (to lie in overfitting region)

**Observations:**

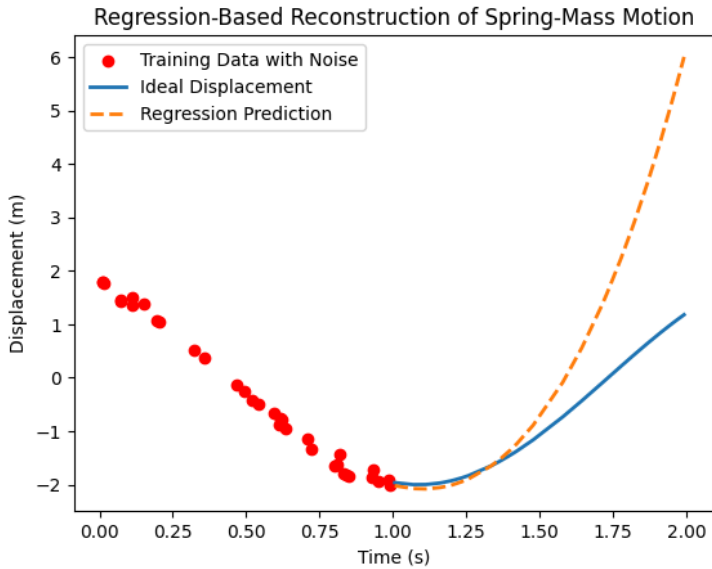
- The training error starts low and more or less maintains the same value, even as data set size increases
- The testing error starts extremely high, but then faces a sharp drop in value and stabilises about the same value as training error as data set size increases
- The model starts becoming more and more accurate as the data set size increases, as indicated by the decrease in testing error

Comparing the Model with and without regularisation:

For degree = 3

And $\lambda = 10^{-3}$ for L2 regularisation

With



Without

